OUTLIERS, Z-SCORE, IQR, NORMAL DISTRIBUTION AND SKEWNESS

1. OUTLIERS

Definition

Outliers are values that are very far from the rest of the data.

Example

Data = [10, 12, 11, 13, 100]
100 is an outlier.

Importance of Outliers

- Distort mean
- Affect analysis
- May represent errors or rare events
- Important in decision making

2. Z-SCORE METHOD (USING MEAN)

Concept

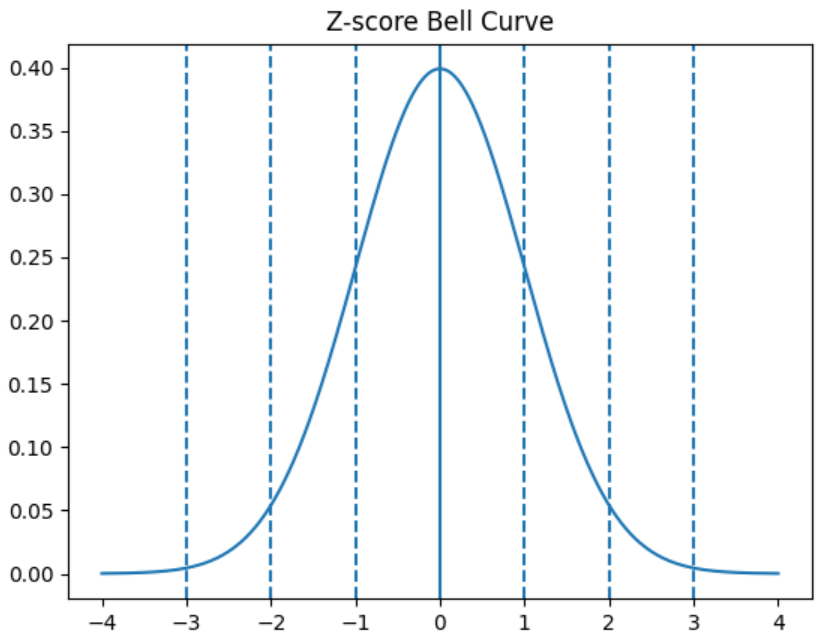
Z-score tells how many standard deviations a value is away from the mean.

$z = (x - \mu) / \sigma$

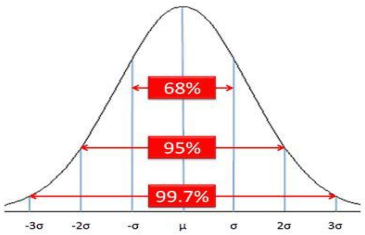
Interpretation Table

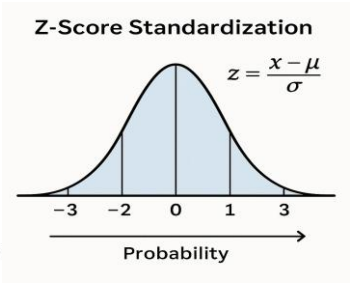
Z-score range	Meaning
0	Exactly at mean
±1	Close to mean
±2	Moderate deviation
±3	Extreme value
>3 or <-3	Outlier

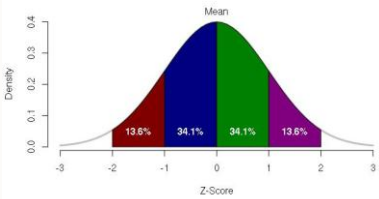
Z-Score Visual (Bell Curve)



Z-Score Visual (Bell Curve)







Normal Distribution Rule

68% within 1 SD
95% within 2 SD
99.7% within 3 SD

Example

Data = [10, 12, 11, 13, 100]
Mean ≈ 29.2
SD ≈ 35
Z-score of 100 ≈ 2.02

Python

```
import numpy as np

data = [10,12,11,13,100]
mean = np.mean(data)
std = np.std(data)
[(x - mean)/std for x in data]
```

Conclusion

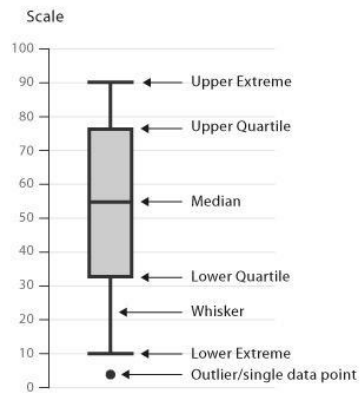
Uses mean and standard deviation. Sensitive to outliers. Best for normal distribution.

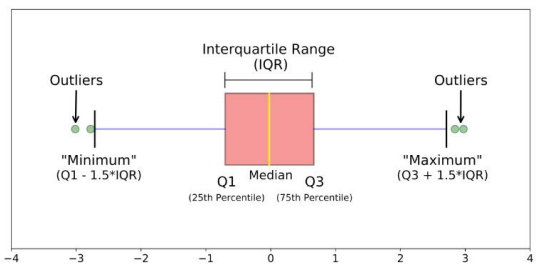
3. IQR METHOD (USING MEDIAN)

Concept

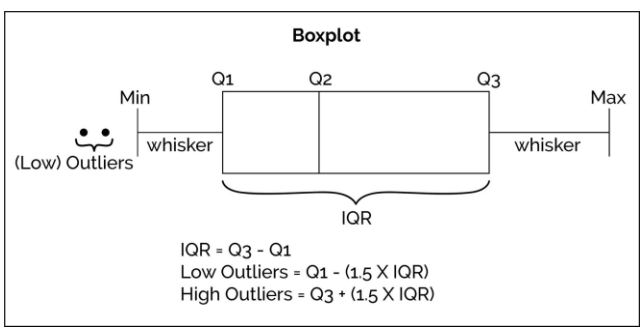
IQR uses quartiles and is not affected by extreme values.
IQR = Q3 - Q1

Boxplot Diagram (IQR)





Different parts of a boxplot



Rule

Lower = Q1 - 1.5×IQR
Upper = Q3 + 1.5×IQR

Example

Data = [10,12,11,13,100]
Q1=11, Q3=13, IQR=2
Lower=8, Upper=16 → 100 is outlier

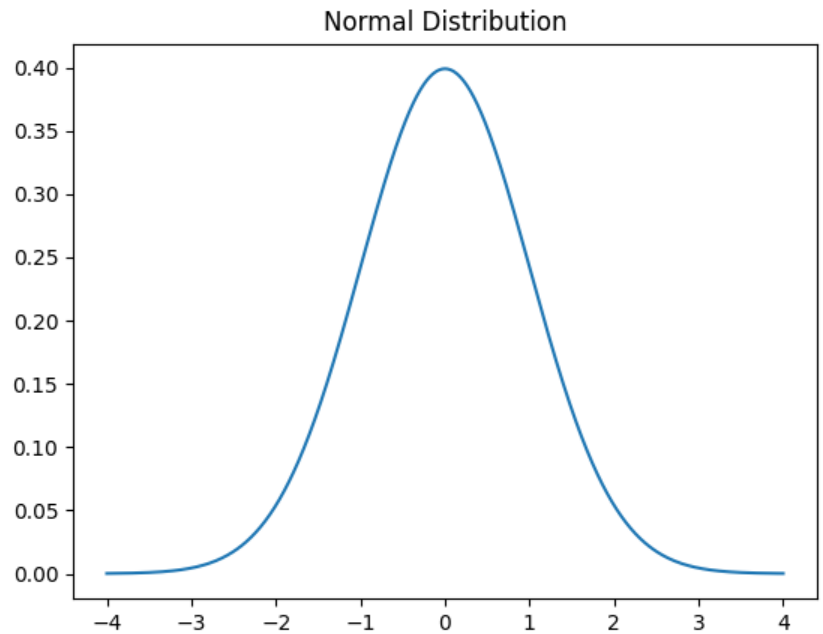
Python

```
import numpy as np
data = [10,12,11,13,100]
Q1 = np.percentile(data,25)
Q3 = np.percentile(data,75)
IQR = Q3 - Q1
[x for x in data if x < Q1 - 1.5*IQR or x > Q3 + 1.5*IQR]
```

Conclusion

Uses median. Not affected by outliers. Most reliable method.

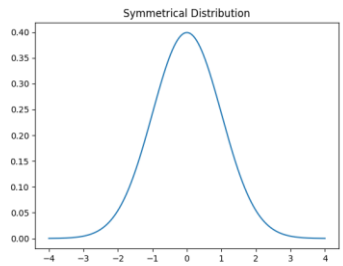
4. NORMAL DISTRIBUTION

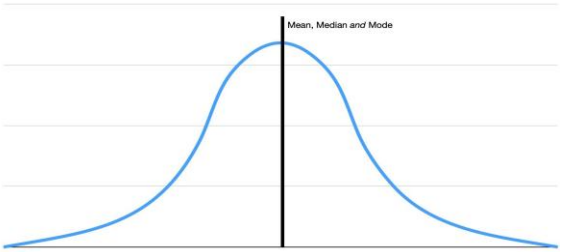


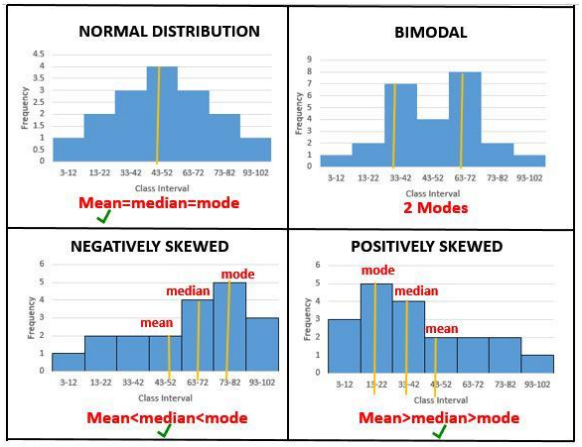
Mean = Median = Mode
Symmetrical bell curve

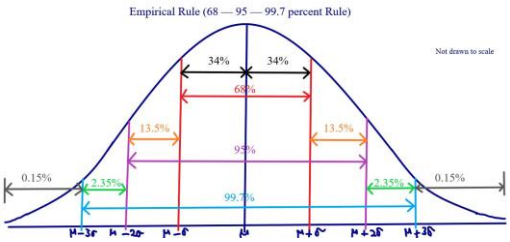
5. SKEWNESS

Symmetrical



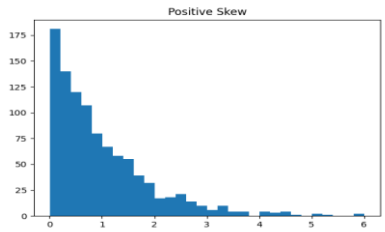


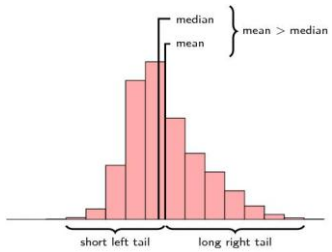


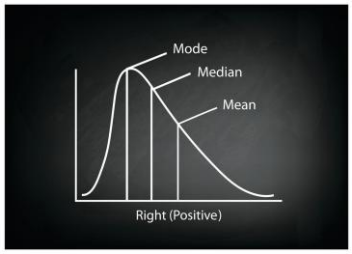


Mean = Median = Mode

Positive Skewness



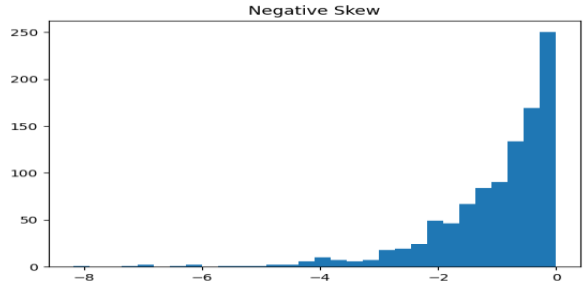


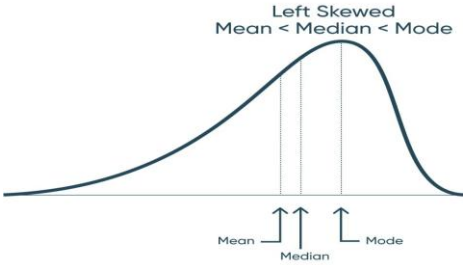


Tail on right side

Mean > Median > Mode

Negative Skewness





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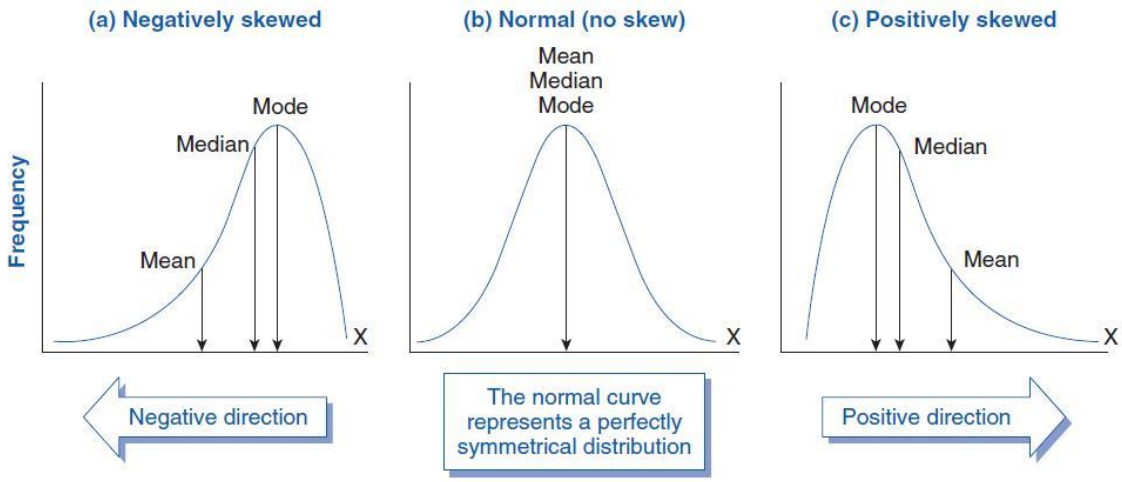
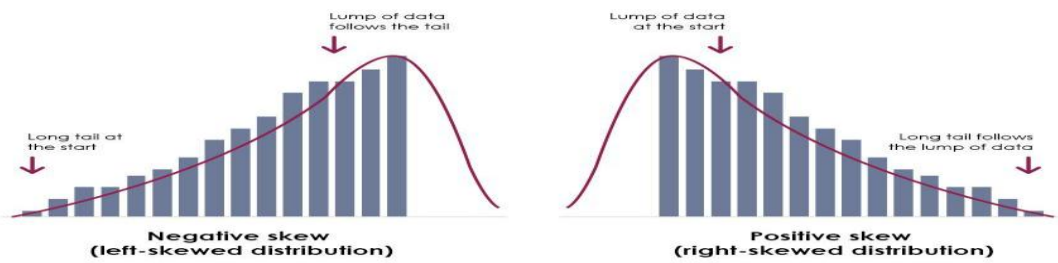
- Tail on left side

Mean < Median < Mode





Note:

- Z-score → based on mean
- IQR → based on median
- Normal distribution → symmetric
- Skewness → asymmetric distribution
- Outliers → extreme values

- Use Z-score for normal data
- Use IQR when outliers exist
- Boxplot helps visualize outliers
- Skewness shows direction of spread